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Proposal

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I confirm that I understand my coursework needs to be submitted online via MST Classroom under the relevant module page before the deadline for my assignment to be accepted and marked. I am fully aware that late submissions will be treated as non-submission and a mark of zero will be awarded.

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1. Introduction

In our chaotic everyday world where life is becoming fast paced, and we as humans are seeking new ways to make our way of life easy and more convenient, one of the most important aspects of our everyday life, being our daily wear is still not up to date to our steady fast lifestyle. We know that clothes make the first impression and are a way to express ourselves, but not everyone of us are born with a sense of style and most of us struggle with making our outfit in a hurry through unorganized closets and hit and trial on matching clothes. Studies suggest that an average person spends about 10-17 minutes per day to choose an outfit, some can decide in seconds while some even take 30 minutes or more (Quora, 2015).

To solve this 'Closet Buddy' comes: an AI powered virtual closet assistant, that can help us organize our wardrobe and create daily outfit for us while also keeping track of our outfits and clothes.

1.1 Problem Scenario

As of 2024 people are buying 60% more clothing items than 15 years ago (McKinsey & Company, 2025). Not only this but there is a decline of 35% to the amount of times a clothing item is worn before thrown away than 15 years ago, where a study suggests that a clothing item is only worn about 7 to a maximum of 10 times before being thrown away (uniformmarket, 2025). The textile and fashion industry annually generates about 92 million tons of textile waste, of which a significant portion ends up in landfills (United Nations Environment Programme (UNEP), 2025).

Many people struggle with their wardrobes and outfit coordination, leading to time inefficiencies while dressing. According to research on a daily basis an individual manages to wear only 20% of the clothes that they own (The Carbon Literacy Trust, 2025) and a survey revealed that 61% of the respondents forget what they own, which results in unnecessary purchases and low reuse rate (Bang, 2022). A person on average feel like they have nothing to wear at least 36 times a year which is termed as "wardrobe panic", especially for big dates or event (HERStyle, 2023).

Traditional wardrobe management is completely manual, unstructured and offers no intelligent solutions to visualize their wardrobe, track usage frequency, and receive intelligent outfit recommendations. This lack of a unified digital platform results in wasted time, disorganized storage and under-utilization of owned clothes.

1.2 Project as a solution.

To solve both the personal and environmental challenges and problems, the proposed project titled “Closet Buddy” aims to develop and design an AI-driven virtual wardrobe application that digitalizes a user’s storage spaces and manages the storage units to better help them organize, understand, and reuse their existing wardrobe to facilitate the users. Key features include:

- Clothing image authentication to organize data.
- Automated segmentation and classification of the uploaded clothing item.
- A virtual storage system (Closets/Drawers/Boxes) in which the users can map their real-life storage spaces digitally.
- Users can view all their pieces and filter by type and usage or view items by storage space.
- A recommendation ai suggests coordinated outfits, helping users make better styling choices
- The system tracks usage frequency of the clothes and prioritizes underused garments to encourages sustainable fashion.
- Integrates with weather to give useful suggestions.

2. Aims and Objectives

Aim:

To develop an AI-powered virtual storage and wardrobe assistant that authenticates and digitizes clothes into a well-structured storage component, recommends daily outfits based on user requirements, promotes sustainability, and integrates with weather Api to enhance user convenience and reduce fast-fashion-driven waste.

Objectives:

1. To provide the users with a user-friendly convenient way to manage and track their real-life storage spaces, specially targeted towards clothing management and keeping record of their outfits.
2. To integrate authentication, classification and outfit recommendation features with AI and provide the users with a service to generate outfits according to their requirements.

3. Expected Outcomes and Deliverables

At the end of this project “Closet Buddy” is expected to deliver the following:

3.1 Software Deliverables:

- A fully functional AI-powered virtual storage management application with following features:
 - User authentication and profile management.
 - Clothing item authentication after uploading image.
 - Clothing segmentation, and classification of clothing metadata.
 - Virtual storage dashboard with all the user storage spaces.
 - Clothing item to be uploaded on specified storage space, can be edited.
 - Storage spaces can also be used to store non-clothing items.
 - Outfit recommendation hybrid engine (Rule-based + AI) that improves with user rating.
 - Clothes inside the storage spaces that are “put away” will not be used for outfit generation.
 - Providing helpful tips and suggestions based on weather.
- Potential Feature:
 - Users can create their own outfit which can be rated by ai to help improve dressing sense.

3.2 Technical deliverables:

- Authentication model to validate clothing item.
- ML pipelines.
- RESTful API architecture.
- Database schema.
- Responsive frontend interface.

3.3 Documentation:

- Test reports and evaluation of models.
- A Project report with system designs, models, work break down structure.

4. Project risks, threats and contingency plans

Table 1: Project risks/threats and contingency plane

S. No.	Risk/Threat	Effect	Impact	Contingency Plan
1.	Segmentation accuracy issues	Poor cutout quality	Medium	Using stronger models, option to proceed without segmentation, asking the user to provide high quality pictures.
2.	Model misclassification	Incorrect tags	High	Fine-tune models with custom dataset, manual correction option, segment images before processing.
3.	Frontend and backend having integration issue	Data loss, loss of functionality.	High	Ensuring smooth integration between backend and frontend.
4.	Data loss	Loss of workable data.	High	Regular backups, cloud storage.
5.	Time constraints for feature completion	Incomplete project	Medium	Prioritize core features, add buffer time.
6.	Users not uploading appropriate images	AI Model will not work as intended	High	Giving clear guidelines and instruction on what type of image to upload, adding validation.
7.	Clothing image authentication error.	Wrong uploads or correct images being rejected	High	Using a better classification model or architecture.
8.	Weather Api malfunctioning	Suggestions can not be made based on weather	Low	Using good quality Apis.

5. Methodology

The **Iterative Incremental Model** was deemed suitable for the methodology of this project because this method helps to break down complex projects into smaller parts instead of figuring out everything at the beginning (geeksforgeeks, 2024). It combines structured development with flexibility allowing the developers to focus on making each part working well before adding more, resulting in a better overall system.

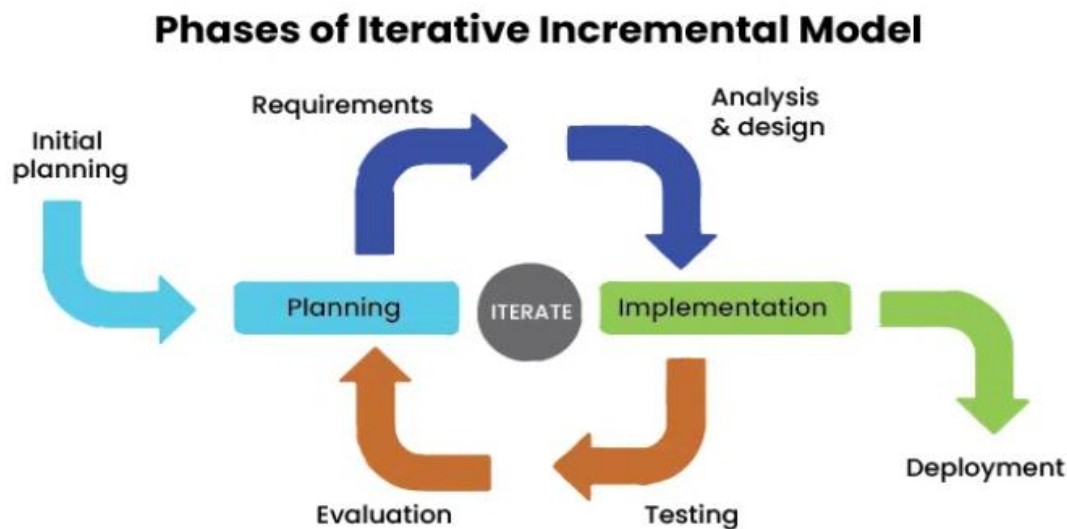


Figure 1: Iterative Incremental Model

The system will be built through small increments, where each increment adds functional components like authentication, segmentation, classification, closet storage, recommendations, etc. Each increment itself goes through iterations of planning, design, implementation, and testing, ensuring continuous refinement and reduced development risk. This model aligns perfectly well with the experimental and modular nature of the project especially in AI-driven components.

5.1 Phases

1) Requirement Analysis and Planning

- Collect functional requirements of the project: segmentation, classification, outfit recommendation, storage units.
- Define the non-functional requirements.
- Prepare datasets for classification.
- Identifying external dependencies.

2) System and Architecture Design

- Architecture Design: Flutter frontend, Django Backend, AI Layer, PostgreSQL database.
- Design Dataflow.
- Design recommendation pipeline.

3) Incremental Development Cycles

The development will be carried out in different incremental development cycles to complete feature on time and for review.

- Increment 1: Authentication module
- Increment 2: Segmentation module
- Increment 3: Classification module
- Increment 4: Storage system development
- Increment 5: Recommendation AI with weather feature integration
- Increment 6: Final system development and integration

4) Testing and Evaluation

- Unit tests for each feature.
- UI testing of Flutter.
- Integration testing for backend and frontend
- Model performance testing.
- Performance testing
- Final system testing

5) Deployment (Not within the project Scope)

- Deploy backend to cloud (AWS)
- Connect Flutter app to deployed API

6) Maintenance and Iteration

- Fixing bugs and refining UI
- Perform optimization based on feedback
- Finalize report and documentation

6. Resource Requirement

6.1 Hardware Requirements

- Good quality laptop with the following specifications:
 - Multi-core processor: Intel i9 or equivalent
 - RAM: 32 GB or higher
 - SSD storage: 512GB or higher
 - GPU: NVIDIA RTX 4050 or equivalent
- A standard mobile phone

6.2 Software Requirements

- IDE and Testing: Visual Studio Code, Postman, Google Colab/ Jupyter Notebook.
- Wireframe: Miro/Canva
- Frontend: Flutter/React Native
- Backend:
 - Python
 - Django
 - TensorFlow/PyTorch/Scikit-learn
 - Hugging face models
- Cloud Storage: AWS
- Version control: GitHub
- Database: PostgreSQL

6.3 Cloud/External Compute Resources

If local hardware is insufficient for heavy AI workload:

- Colab T4 GPU/ v5e-1 TPU

6.4 Datasets

- <https://github.com/rainalexotl/deepfashion-image-classification>

7. Work Break Down Structure

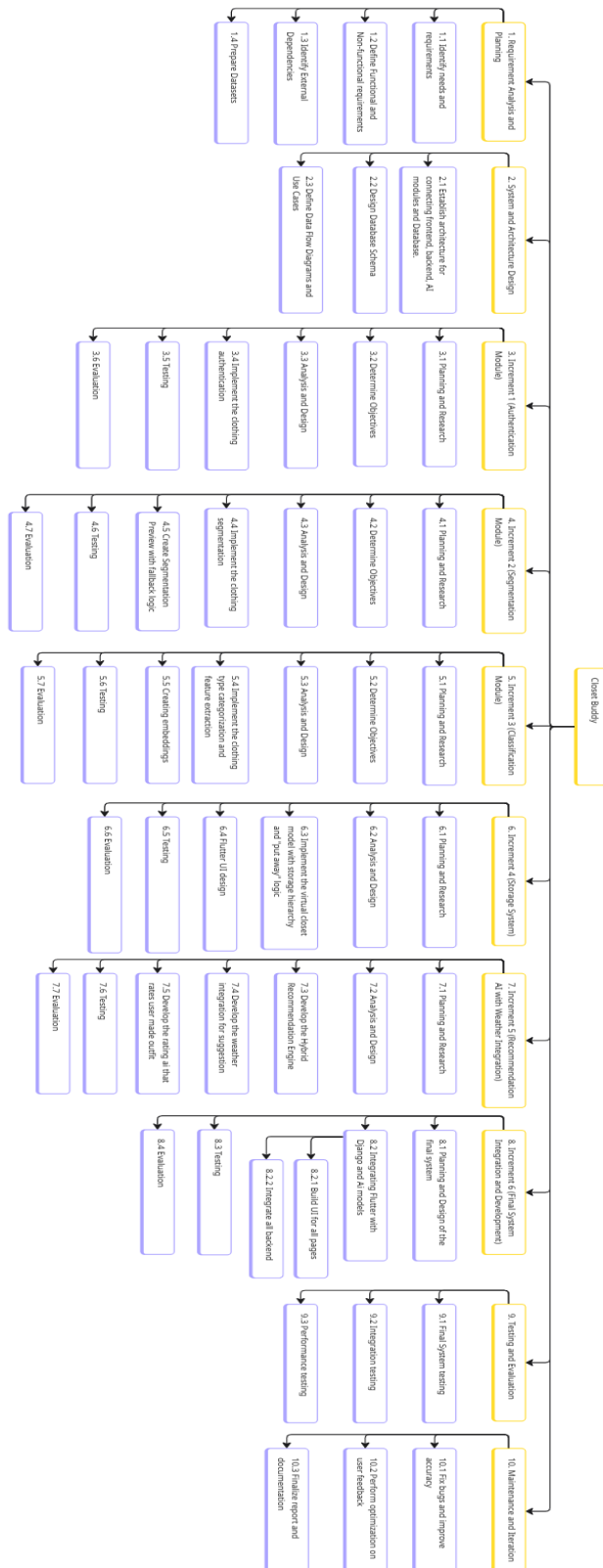


Figure 2: WBS

Partitioning the figure for clarity, WBS Part-1:

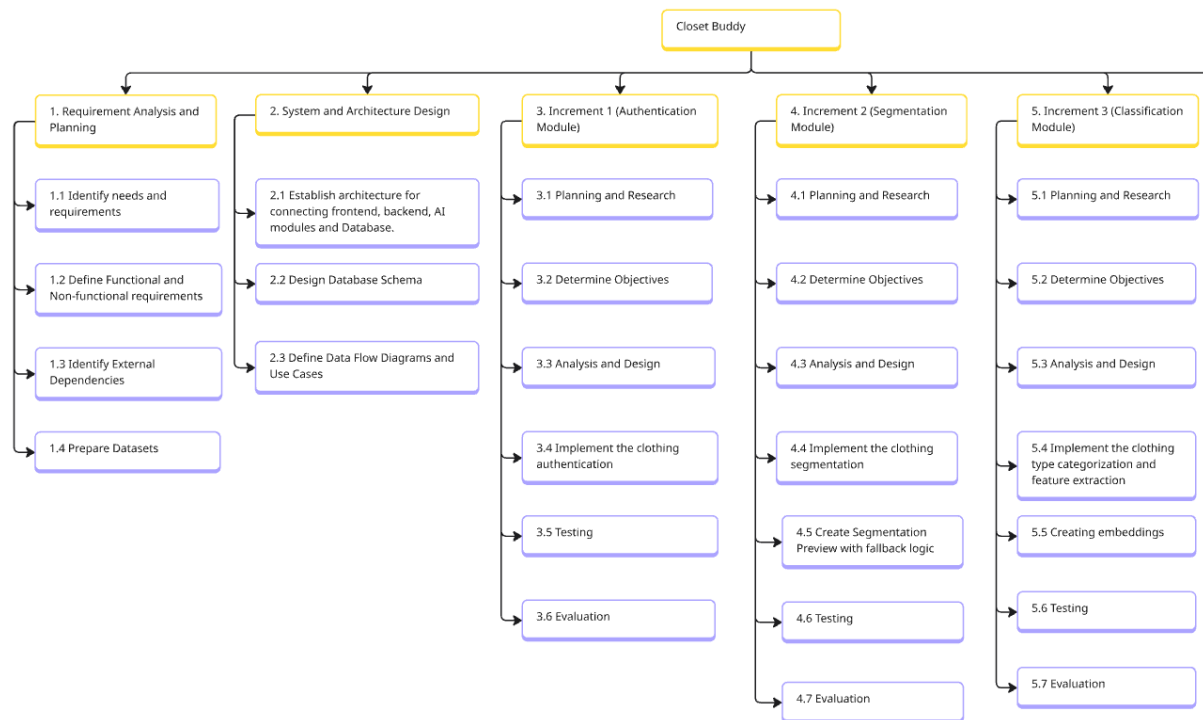


Figure 3: WBS Part-1

WBS Part-2:

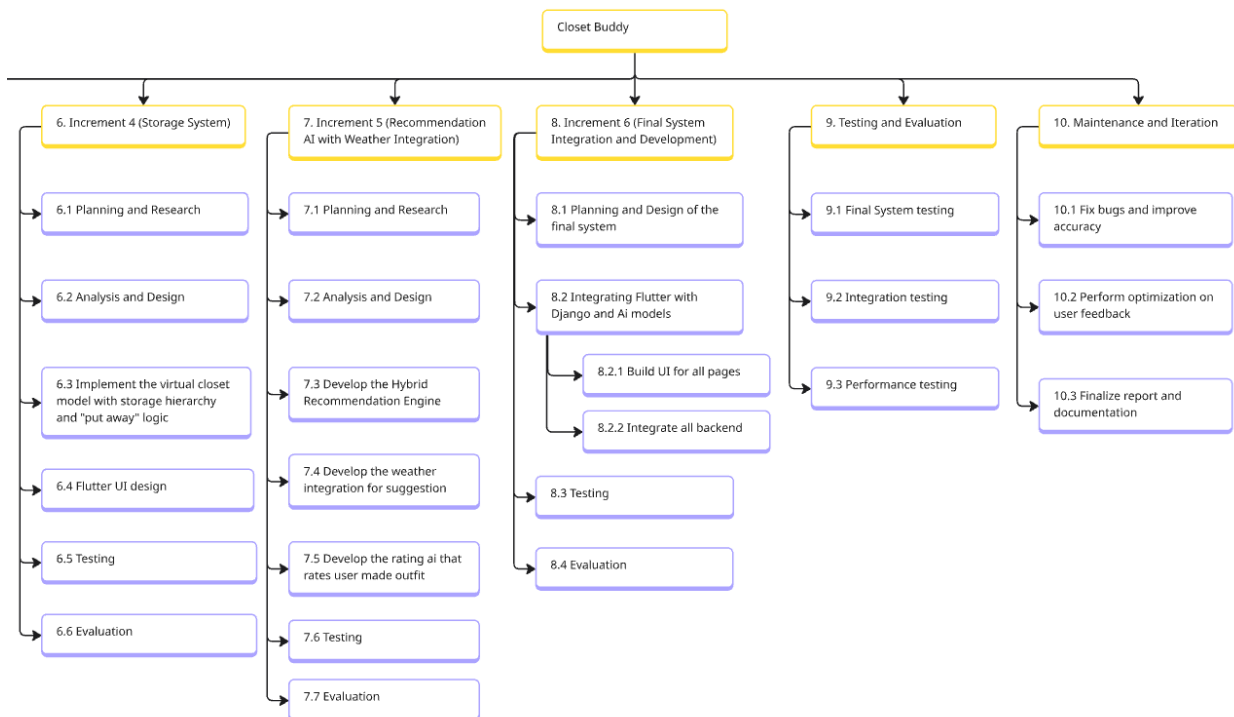


Figure 4: WBS Part-2

8. Milestones

Table 2: Milestone Table

S. No.	Milestone	Target Date	Description
1.	Topic Finalization	November 3, 2025	"Closet Buddy" was accepted as the FYP topic.
2.	Proposal Submission	December 3, 2025	A detailed project proposal was submitted.
3.	Authentication and Segmentation functional	December 21, 2025	The uploaded image is successfully authenticated and segmented
4.	Classification and meta data extraction successful	December 31, 2025	Successfully extracted clothing meta data like type, category, colour, and made embeddings.
5.	Completed the storage interface	January 10, 2025	Clothing items are stored in database with metadata and storage units front end completed. Items are displayed in frontend.
6.	Recommendation engine functional	January 18, 2025	Recommendation engine works and generates outfit.
7.	Weather Integration and AI rating complete.	January 25, 2025	Gives suggestion based on weather and users can get the outfit they created rated by AI.

8.	Backend Frontend Integration Completed	February 15, 2025	Full backend complete & stable, flutter UI integrated with backend.
9.	Testing Phase Done	March 25, 2025	Testing Completed
10.	Final Report Submission	April, 2025	Final report submitted.

9. Gantt Chart

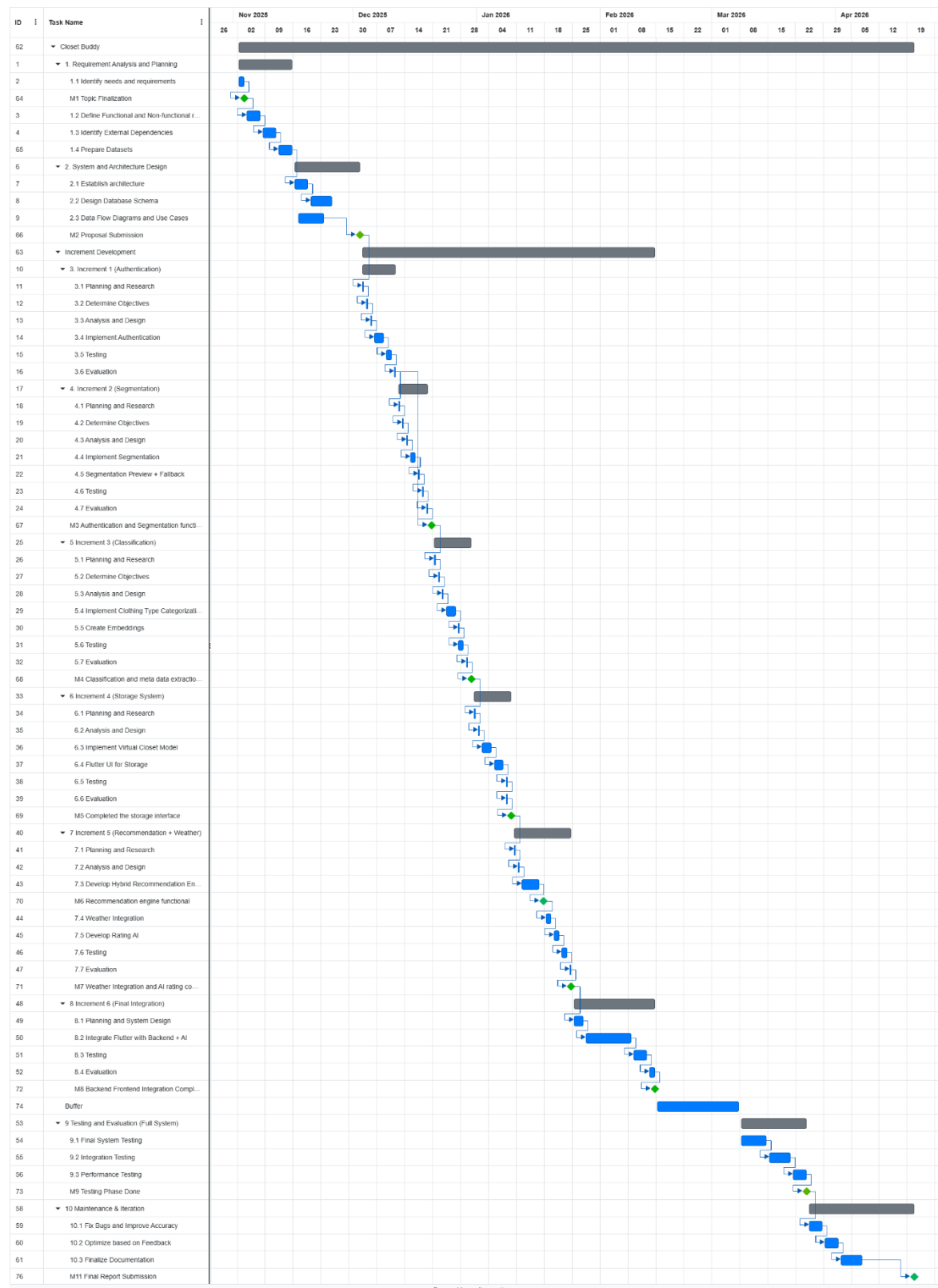


Figure 5: Gantt Chart

[Click here to see the task list and its details](#)

10. Conclusion

The project involves the design and development of Closet Buddy, an AI-powered wardrobe and storage management application with outfit recommendation system that will fulfil user needs involving the digital organization of one's closet and styling of their outfits. The platform integrates clothing authentication, segmentation, metadata extraction, a virtual grid-based closet, hybrid outfit generation, and weather-aware suggestions into a single powerful application. Designed for ease and convenience, the system emphasizes accessibility, usability, and adaptability across a wide range of users.

Care was taken in the planning, iterative design, and systematic testing to realize reliable performance and accurate AI-driven recommendations. The main focus is the hybrid outfit generation engine that combines both rule based and AI implementation methods for better performance along with a flexible virtual storage model and integration of interactive features that allow for efficient utilization of the wardrobe.

The end result of this project is a user friendly and easy to use centralized application for storage management and digitalization, that not only helps for generating outfit ideas and therefore encouraging the users to actively engage with their already owned clothing collection in a more fun and creative manner while also helping to reduce textile waste and being a personal assistant for the closet. As a bonus it also helps keep our storage spaces organized, mapped, and easy to keep record of.

11. Bibliography

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12. Appendix

12.1 Gantt Chart Task Table

Table 3: Gantt Chart Task Table

Task Name	Start	Finish	Duration
1. Requirement Analysis and Planning	11/2/2025	11/15/2025	14 day
1.1 Identify needs and requirements	11/2/2025	11/3/2025	2 day
M1 Topic Finalization	11/3/2025	11/3/2025	Milestone
1.2 Define Functional and Non-functional requirements	11/4/2025	11/7/2025	4 day
1.3 Identify External Dependencies	11/8/2025	11/11/2025	4 day
1.4 Prepare Datasets	11/12/2025	11/15/2025	4 day
2. System and Architecture Design	11/16/2025	12/2/2025	17 day
2.1 Establish architecture	11/16/2025	11/19/2025	4 day
2.2 Design Database Schema	11/20/2025	11/25/2025	6 day
2.3 Data Flow Diagrams and Use Cases	11/17/2025	11/23/2025	7 day
M2 Proposal Submission	12/2/2025	12/2/2025	Milestone
Increment Development	12/3/2025	2/14/2026	74 day
3. Increment 1 (Authentication)	12/3/2025	12/11/2025	9 day
3.1 Planning and Research	12/3/2025	12/3/2025	1 day
3.2 Determine Objectives	12/4/2025	12/4/2025	1 day
3.3 Analysis and Design	12/5/2025	12/5/2025	1 day
3.4 Implement Authentication	12/6/2025	12/8/2025	3 day
3.5 Testing	12/9/2025	12/10/2025	2 day
3.6 Evaluation	12/11/2025	12/11/2025	1 day
4. Increment 2 (Segmentation)	12/12/2025	12/19/2025	8 day
4.1 Planning and Research	12/12/2025	12/12/2025	1 day
4.2 Determine Objectives	12/13/2025	12/13/2025	1 day
4.3 Analysis and Design	12/14/2025	12/14/2025	1 day
4.4 Implement Segmentation	12/15/2025	12/16/2025	2 day
4.5 Segmentation Preview + Fallback	12/17/2025	12/17/2025	1 day
4.6 Testing	12/18/2025	12/18/2025	1 day
4.7 Evaluation	12/19/2025	12/19/2025	1 day

M3 Authentication and Segmentation functional	12/20/2025	12/20/2025	Milestone
5. Increment 3 (Classification)	12/21/2025	12/30/2025	10 day
5.1 Planning and Research	12/21/2025	12/21/2025	1 day
5.2 Determine Objectives	12/22/2025	12/22/2025	1 day
5.3 Analysis and Design	12/23/2025	12/23/2025	1 day
5.4 Implement Clothing Type Categorization	12/24/2025	12/26/2025	3 day
5.5 Create Embeddings	12/27/2025	12/27/2025	1 day
5.6 Testing	12/27/2025	12/28/2025	2 day
5.7 Evaluation	12/29/2025	12/29/2025	1 day
M4 Classification and meta data extraction successful	12/30/2025	12/30/2025	Milestone
6. Increment 4 (Storage System)	12/31/2025	1/9/2026	10 day
6.1 Planning and Research	12/31/2025	12/31/2025	1 day
6.2 Analysis and Design	1/1/2026	1/1/2026	1 day
6.3 Implement Virtual Closet Model	1/2/2026	1/4/2026	3 day
6.4 Flutter UI for Storage	1/5/2026	1/7/2026	3 day
6.5 Testing	1/8/2026	1/8/2026	1 day
6.6 Evaluation	1/8/2026	1/8/2026	1 day
M5 Completed the storage interface	1/9/2026	1/9/2026	Milestone
7. Increment 5 (Recommendation + Weather)	1/10/2026	1/24/2026	15 day
7.1 Planning and Research	1/10/2026	1/10/2026	1 day
7.2 Analysis and Design	1/11/2026	1/11/2026	1 day
7.3 Develop Hybrid Recommendation Engine	1/12/2026	1/16/2026	5 day
M6 Recommendation engine functional	1/17/2026	1/17/2026	Milestone
7.4 Weather Integration	1/18/2026	1/19/2026	2 day
7.5 Develop Rating AI	1/20/2026	1/21/2026	2 day
7.6 Testing	1/22/2026	1/23/2026	2 day
7.7 Evaluation	1/24/2026	1/24/2026	1 day
M7 Weather Integration and AI rating completed	1/24/2026	1/24/2026	Milestone

8. Increment 6 (Final Integration)	1/25/2026	2/14/2026	21 day
8.1 Planning and System Design	1/25/2026	1/27/2026	3 day
8.2 Integrate Flutter with Backend + AI	1/28/2026	2/8/2026	12 day
8.3 Testing	2/9/2026	2/12/2026	4 day
8.4 Evaluation	2/13/2026	2/14/2026	2 day
M8 Backend Frontend Integration Completed	2/14/2026	2/14/2026	Milestone
Buffer	2/15/2026	3/7/2026	21 day
9. Testing and Evaluation (Full System)	3/8/2026	3/31/2026	24 day
9.1 Final System Testing	3/8/2026	3/14/2026	4 day
9.2 Integration Testing	3/15/2026	3/20/2026	4 day
9.3 Performance Testing	3/21/2026	3/24/2026	4 day
M9 Testing Phase Done	3/25/2026	3/25/2026	Milestone
10. Maintenance & Iteration	3/25/2026	4/21/2026	21 day
10.1 Fix Bugs and Improve Accuracy	3/25/2026	3/28/2026	4 day
10.2 Optimize based on Feedback	3/29/2026	4/1/2026	4 day
10.3 Finalize Documentation	4/2/2026	4/07/2026	6 day
M10 Final Report Submission	4/21/2026	4/21/2026	Milestone

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